

LIBRARY MANAGEMENT SYSTEM

Project Report

Applicant: Jobanputra Yash Shaileshbhai

Application ID: KTS020260903193

Internship Domain: Python Development Internship

Submitted as part of the Python Development Internship project requirements.

1. Introduction

The Library Management System is a simple application developed to manage books, library members and book issue/return records. The system provides an easy way to store and manage library information using a database.

2. Objectives

- Manage book information
- Manage member information
- Issue books to members
- Return issued books
- View issue and return records
- Store information using SQLite
- Provide input validation and basic exception handling

3. Technologies Used

Python: Main programming language.

Tkinter: Used to create the desktop graphical user interface.

Flask: Used for the deployed web version.

SQLite: Used for database storage.

HTML/CSS: Used for the web interface.

GitHub: Used for source-code hosting.

Render: Used for live deployment of the web version.

4. Main Features

Book Management: Add and delete books and maintain quantity and available-book information.

Member Management: Add and delete library members with contact details.

Issue & Return: Issue available books to members and record their return.

Search & Filter: Search/filter books using relevant information in the web version.

Records: View issue and return history.

5. Database

The system uses SQLite for storing application data. The database keeps records related to books, members and issue/return transactions. The database is created automatically when the application is run.

6. Working of the System

The user first adds books and members. When a book is issued, the system checks its availability and creates an issue record. The available quantity is reduced. When the book is returned, the record is updated and the available quantity is increased. This keeps the library records organized.

7. Web Deployment

A Flask-based web version of the Library Management System was created separately from the Tkinter desktop application. The web version was deployed online and provides access to the main library management functions through a browser.

Live Project: <https://library-management-system-c3g3.onrender.com>

8. Project Repository

The complete source code, README documentation and deployment files are maintained in the public GitHub repository.

9. Conclusion

The Library Management System successfully demonstrates basic library operations such as book management, member management and issue/return tracking. The project also provided practical experience with Python, SQLite, Flask, GitHub and web deployment.

10. Project Screenshots

Library management system

Library Management System

Manage books, members and issue / return records

BooksMembersIssue / ReturnRecords

Add New Book

Book Name

Author

Category

Quantity

Add Book

ID	Book Name	Author	Category	Quantity	Available
1	python dev	yash	developer	20	20
2	python basic	r.sharma	programming	3	2
3	java advanced	ravi kishan	programming	5	5

Delete Selected Book

Library Management System

Manage books, members and issue / return records

BooksMembersIssue / ReturnRecords

Issue / Return Records

Issue ID	Book	Member	Issue Date	Return Date	Status
3	python basic	yah	2026-10-04	None	Issued
2	python basic	yah	2026-10-04	None	Issued
1	python dev	yah	2026-10-04	2026-10-04	Returned

Refresh Records

Library management system

Library Management System

Manage books, members and issue / return records

BooksMembersIssue / ReturnRecords

Add New Member

Name

Phone

Email

Add Member

	ID	Name	Phone	Email
1		yah	9925569704	yash@gmail.com
2		raju	9276367296	raju12@outlook.com

Delete Selected Member

Library Management System

Library Management System

Manage books, members and issue / return records

BooksMembersIssue / ReturnRecords

Issue or Return a Book

Issue Book

Book ID

Member ID

Issue Book

Return Book

Issue ID

Return Book

Use the Book ID and Member ID from the Books and Members tabs.